

Invoice Portal Admin: Architecture

Source of record: `ARCHITECTURE.md` in the repo root Generated 2026-09-17 by `docs/build.js`

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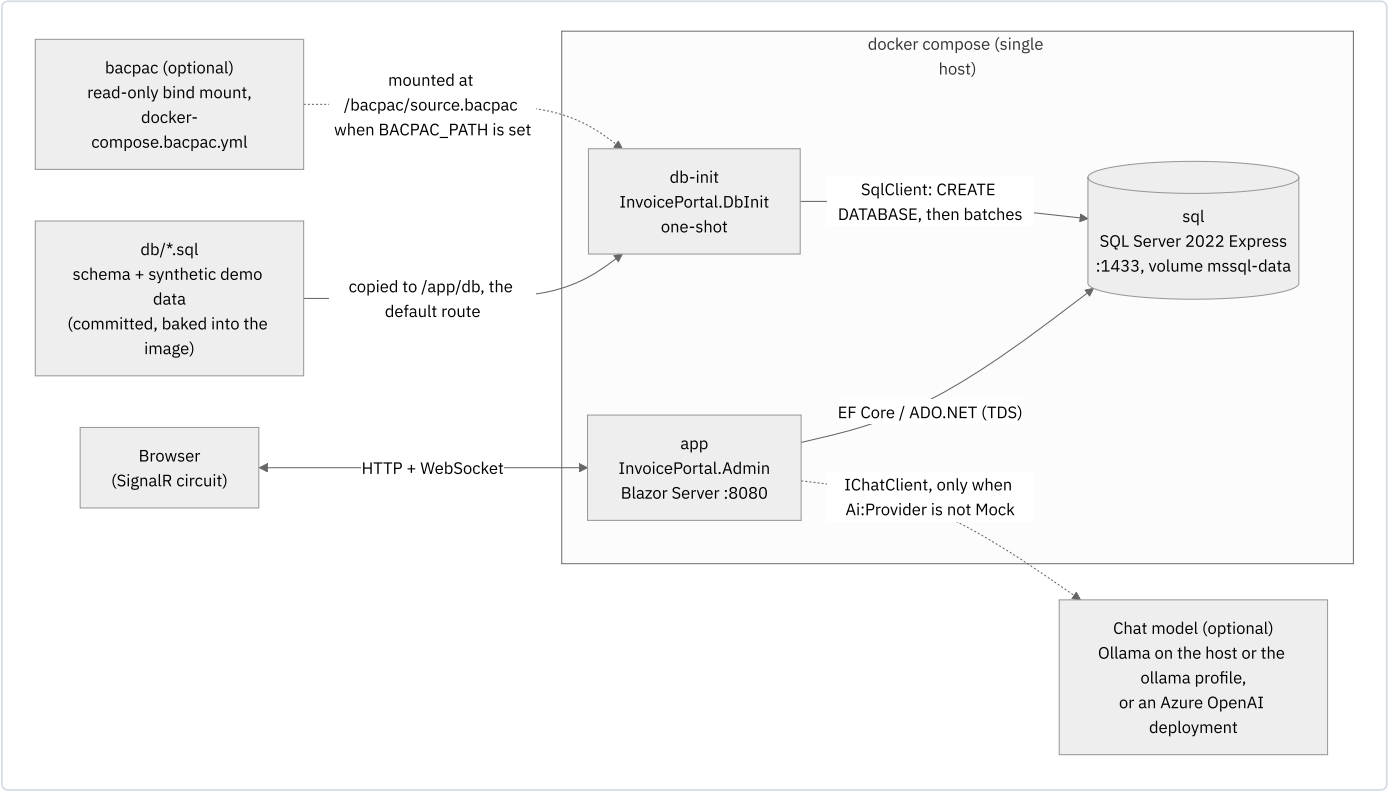
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This document describes the structure, data flows and integration points of the Invoice Portal Admin proof of concept. It complements README.md, which covers how to run the solution.

The diagrams are Mermaid. If your Markdown viewer does not draw them, open docs/ARCHITECTURE.html in a browser: it is the same content with the diagrams rendered (needs network access once to load mermaid.js from a CDN). A PDF is at docs/ARCHITECTURE.pdf. Both are generated from this file; after editing, rebuild them with `npm run build:pdf` in the `docs/` folder (see `docs/package.json`).

One-paragraph summary. A .NET 10 Blazor Web App (Interactive Server render mode, Radzen components) administers a local copy of the Invoice Portal database. The database is SQL Server 2022 Express running in Docker, built once by a one-shot console app from SQL scripts committed under `db/` - the schema plus synthetic demo data, so the repository needs no production export. All data access goes through an EF Core scaffolded model behind a `DbContext` factory. Two AI features, natural-language-to-SQL and grounded document Q&A, are built on the `Microsoft.Extensions.AI` abstractions. The chat model is pluggable: the default `Mock` provider is an in-process fake that needs no network, `Ollama` talks to a local model, and `AzureOpenAI` talks to an Azure OpenAI deployment with Entra ID credentials. The document index is always in-process. An `azd` template under `infra/` can host the container in Azure Container Apps against the existing Azure SQL database. With the defaults, nothing leaves the machine.

1 System context



Compose start-up ordering is enforced with health conditions: `db-init` waits for `sql` to pass its `sqlcmd` health check, and `app` waits for `db-init` to exit successfully.

SERVICE	IMAGE / BUILD	ROLE
sql	mcr.microsoft.com/mssql/server:2022-latest , MSSQL_PID=Express	Database engine. Data persists in the <code>mssql-data</code> volume.
db-init	InvoicePortal.DbInit/Dockerfile	Builds the database from <code>db/*.sql</code> if it does not exist, or imports a bacpac when <code>BACPAC_PATH</code> is set.
app	InvoicePortal.Admin/Dockerfile	The admin UI. HTTP only, port 8080. Exposes <code>/healthz</code> .
ollama	ollama/ollama:latest , profile <code>ollama</code> only	Optional CPU-only local model. Started with <code>--profile ollama</code> ; data in the <code>ollama-data</code> volume.
aspire-dashboard	mcr.microsoft.com/dotnet/aspire-dashboard:latest , profile <code>otel</code> only	Optional in-memory OTLP sink with a UI on 18888 (see 3.4). Started with <code>--profile otel</code> .

Locally there is no authentication or authorization layer in the app. It is intended for a single developer on a local machine. The Azure deployment (section 8) can put Entra ID sign-in in front of the container with Easy Auth, without application code changes.

2 Solution structure

```
InvoicePortal.slnx
├─ db/
├─ InvoicePortal.DbInit/
├─ InvoicePortal.Admin/
│   └─ Program.cs
│   └─ Components/
│   └─ Services/
│   └─ Logging/
│   └─ Data/
│   └─ Ai/
Documents (RAG)
├─ Telemetry/
Meter
├─ InvoicePortal.Admin.Tests/
├─ .vscode/
├─ .env.example
├─ docs/
├─ infra/
Insights, Azure OpenAI)
├─ azure.yaml
├─ .github/workflows/
deploy)
```

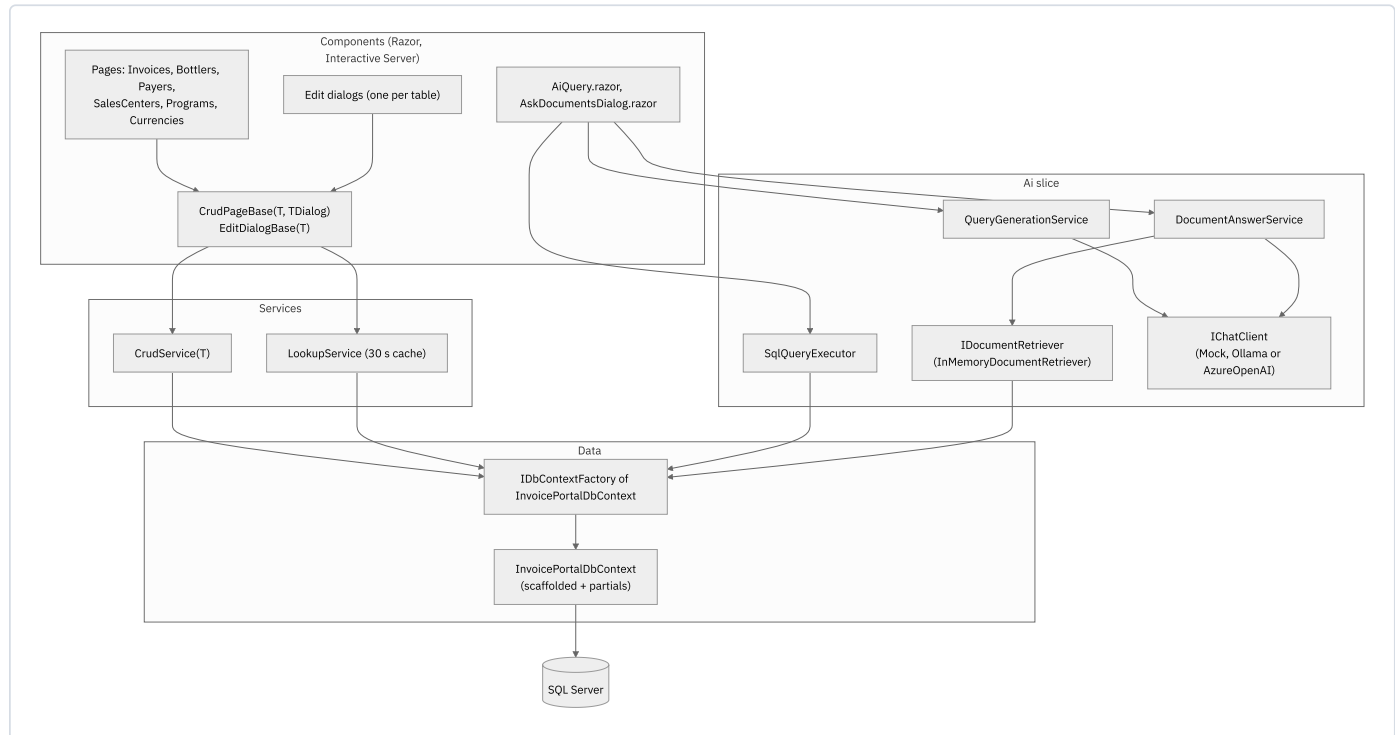
committed database: 001_schema.sql, 002_seed_demo_data.sql, README.md
console app: SQL script runner + bacpac sanitiser/DacFx importer
Blazor Web App
composition root (EntryPoint.Main)
Razor UI: Layout, Pages (Invoices, Lookups, Ai), Shared bases
CrudService<T>, LookupService, EnumDisplay
Serilog routing, validated settings, App Insights correlation, request logging
EF Core scaffold (never hand-edited) + Partial + Enums + Abstractions
AI slice: options, DI, Chat (mock + transport wrapper), Query (NL->SQL),
OpenTelemetry options, provider/exporter wiring, the app's ActivitySource and
xUnit tests for AI, telemetry and logging, no database required
recommended extensions, build/compose tasks, F5 debug configurations
tracked template; copy to .env before the first compose up
this document's rendered copy + build script, Azure deployment guide
Bicep for the Azure deployment (Container Apps, ACR, identity, Blob, App
Azure Developer CLI (azd) service definition
ci.yml (build, test, Bicep validation) and azure-dev.yml (azd provision +
deploy)

Key packages:

PACKAGE	VERSION	USED FOR
Microsoft.EntityFrameworkCore.SqlServer	10.0.12	Data access
Radzen.Blazor	11.4.0	Grids, dialogs, forms, notifications
Microsoft.Extensions.AI (+ Abstractions)	10.10.0	IChatClient seam, pipeline builder, logging decorator
OllamaSharp	5.4.30	IChatClient for a local Ollama server (Ollama provider)
Azure.AI.OpenAI , Microsoft.Extensions.AI.OpenAI	2.1.0, 10.10.0	IChatClient for Azure OpenAI (AzureOpenAI provider)
Azure.Identity	1.21.0	DefaultAzureCredential for Azure OpenAI and Blob Storage
Azure.Extensions.AspNetCore.DataProtection.Blobs	1.5.4	Data Protection key ring in Blob Storage when hosted in Azure
OpenTelemetry.Extensions.Hosting + Instrumentation.AspNetCore , .Http , .SqlClient , .Runtime	1.18.0	Traces, metrics and logs (section 3.4)
OpenTelemetry.Exporter.OpenTelemetryProtocol	1.18.0	OTLP export to the Aspire Dashboard or any collector
Azure.Monitor.OpenTelemetry.Exporter	1.9.0	Application Insights traces and metrics (not logs)
Serilog.AspNetCore	10.0.0	Unified ILogger pipeline, console, bootstrap and shutdown
Serilog.Sinks.File	7.0.0	Local daily/size-rolled JSON logs with retention
Serilog.Sinks.ApplicationInsights	5.0.1	Azure application logs, correlated with OpenTelemetry activities
Serilog.Sinks.Datadog.Logs	0.6.1	Azure logs to Datadog over HTTPS, bounded asynchronous queue
Microsoft.SqlServer.DacFx	170.5.76	Bacpac import (DbInit only)

3 Runtime architecture of the admin app

3.1 Layers



3.2 Dependency injection and lifetimes

Blazor Server keeps one DI scope per SignalR circuit, so "scoped" here means "per browser session".

REGISTRATION	LIFETIME	NOTES
<code>IDbContextFactory<InvoicePortalDbContext></code>	Singleton factory	Contexts are created per operation and disposed immediately. <code>EnableRetryOnFailure(3)</code> .
<code>CrudService<T></code> (open generic)	Scoped	One per entity type per circuit.
<code>LookupService</code>	Scoped	Dropdown sources cached per circuit for 30 seconds.
<code>IChatClient</code>	Singleton	Provider-dependent pipeline chosen by <code>Ai:Provider</code> (see 7.2). Always ends in <code>.UseOpenTelemetry()</code> then <code>.UseLogging()</code> .
OpenTelemetry providers	Singleton	<code>AddOpenTelemetry()</code> with tracing, metrics and logging (see 3.4). Skipped entirely when <code>Telemetry.Enabled</code> is false.
Serilog logger, <code>AppLoggingOptions</code>	Singleton	Logging is configured before telemetry. Clears default logging providers and forwards once to the optional OTLP provider. Independent of <code>Telemetry.Enabled</code> .
<code>IDocumentRetriever</code>	Singleton	<code>InMemoryDocumentRetriever</code> for every provider. Index built lazily on first use.
Health checks	n/a	<code>AddHealthChecks()</code> mapped at <code>/healthz</code> for container probes.
Data Protection	Singleton	Only when <code>DataProtection.BlobUri</code> is set: key ring persisted to Blob Storage with <code>DefaultAzureCredential</code> , so antiforgery tokens survive container restarts. Locally the default file store is used.
<code>SchemaDescriber</code>	Singleton	Schema text derived from EF model metadata, cached.
<code>AiRateLimiter</code>	Scoped	Fixed window, default 10 requests per minute per circuit.
<code>IQueryGenerationService</code> , <code>ISqlQueryExecutor</code> , <code>IDocumentAnswerService</code>	Scoped	Stateless orchestration.
Radzen <code>DialogService</code> , <code>NotificationService</code>	Scoped	Via <code>AddRadzenComponents()</code> .

A factory is used instead of a scoped `DbContext` because a circuit can live for hours. A long-lived tracked context would accumulate state and hold a connection open.

3.3 Shared CRUD behaviour

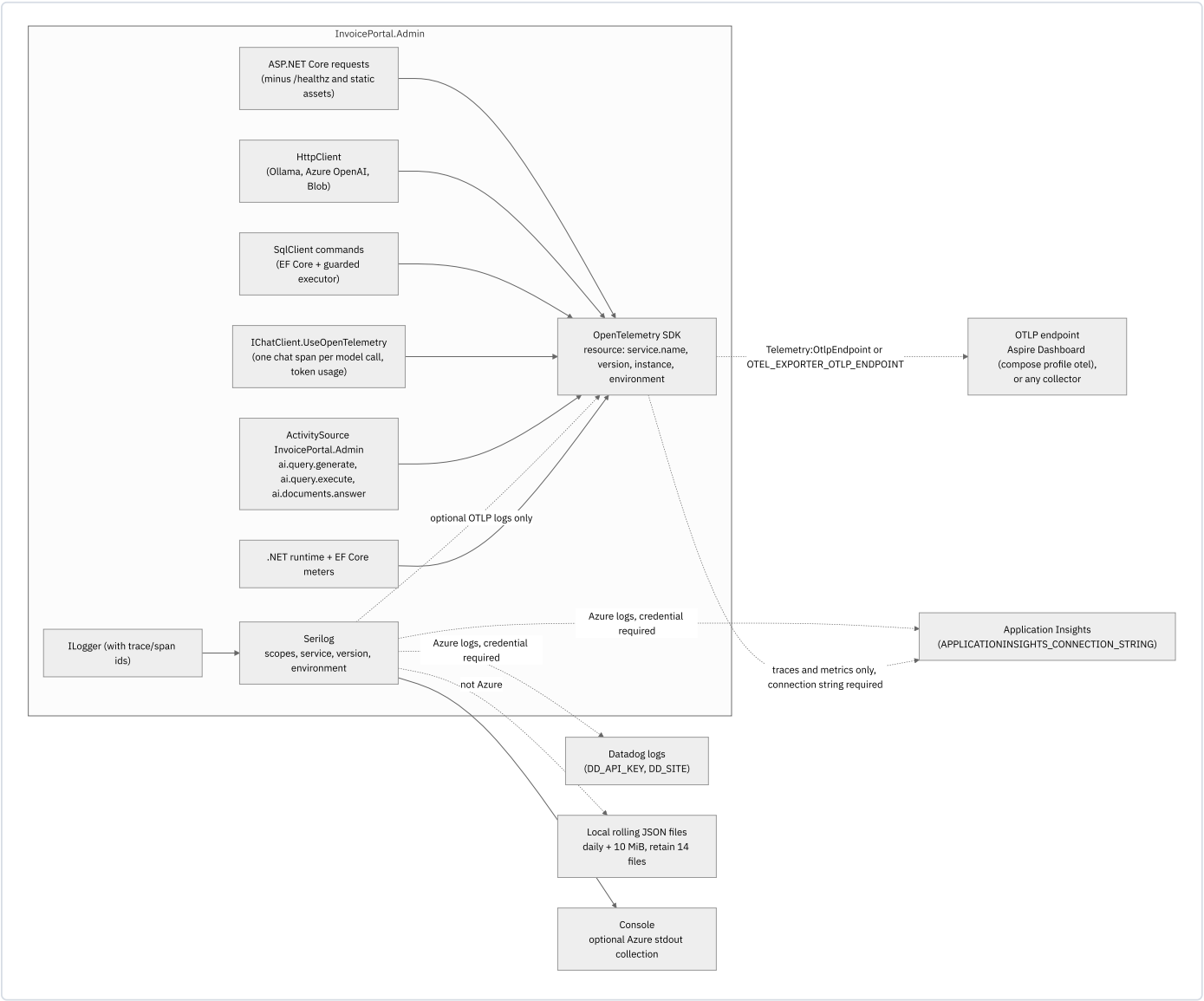
Every grid page inherits `CrudPageBase<TEntity, TDialog>` and every edit dialog inherits `EditDialogBase<TEntity>`. Pages supply only the Radzen markup and a few overrides (`EntityName`, `KeyOf`, `Shape` for `Includes`, `DialogWidth`). The bases provide server-side `LoadData`, Add/Edit through a dialog, Delete with confirmation, Restore, the "Show deleted" toggle, and notifications.

3.4 Observability (Serilog and OpenTelemetry)

`Logging/LoggingExtensions.cs` makes Serilog the sole application logging pipeline. Local hosts (including Docker Production) write console and daily/size-rolled JSON files. Azure hosts write console plus Application Insights and Datadog when their credentials are supplied, without local files. `EntryPoint.Main` uses a console bootstrap logger and flushes on exit. `Telemetry/TelemetryExtensions.cs` keeps OpenTelemetry traces and metrics, plus optional OTLP logs for Aspire. OpenTelemetry can be disabled independently.

Serilog and OpenTelemetry are both registered as `Microsoft.Extensions.Logging` providers: `builder.Logging.ClearProviders()` then `AddSerilog(...)`, and `otel.WithLogging(...)` adds the OTLP provider alongside it. A record written through `ILogger<T>` therefore reaches both - Serilog's sinks and the OTLP exporter. One written straight to Serilog's own API reaches only Serilog's sinks, which is why `Logging/RequestLoggingMiddleware.cs` logs per-

request lines through `ILogger<T>` rather than using `UseSerilogRequestLogging` : the latter would never appear in the Aspire dashboard. That middleware runs first in the pipeline, reports the status the client received, reuses `TelemetryExtensions.IsNoiseRequest` so logs and traces agree on what a real request is, and is switched off with `AppLogging:RequestLoggingEnabled` .



SIGNAL	SOURCES
Traces	ASP.NET Core (filtered by <code>TelemetryExtensions.IsNoiseRequest</code>), <code>HttpClient</code> , <code>SqlClient</code> , <code>InvoicePortal.Admin.Ai</code> (the <code>Microsoft.Extensions.AI</code> decorator, GenAI semantic conventions), <code>InvoicePortal.Admin</code> (own spans), plus the spans <code>.NET 10 Blazor</code> emits itself (<code>Event onclick -> ...</code> , <code>Route ...</code> , <code>Circuit ...</code>). Head sampling <code>Telemetry:TraceSamplingRatio</code> , parent-based. <code>NoiseFilteringProcessor</code> un-records the per-hub-call plumbing spans (<code>ComponentHub/OnRenderCompleted</code> , <code>EndInvokeJSFromDotNet</code> , ...) that would otherwise produce dozens of one-span traces per click; it is a processor, not a sampler, because SignalR names the span only after it starts.
Metrics	ASP.NET Core and Kestrel, <code>HttpClient</code> , <code>.NET runtime</code> , <code>Microsoft.EntityFrameworkCore</code> , chat token usage and duration, <code>invoiceportal.ai.requests</code> (by <code>ai.feature</code> and <code>ai.outcome</code>), <code>invoiceportal.ai.query.rows</code> .
Logs	<code>ILogger</code> and direct <code>Serilog</code> messages, filtered by <code>Serilog</code> levels, with scopes and captured Activity IDs. Console stays on by default. Application Insights uses the captured trace/span IDs and matching role name for correlation.

OpenTelemetry exporters: OTLP (gRPC by default, `http/protobuf` optional) and Azure Monitor traces/metrics, each added only when its endpoint/connection string is present. Serilog is the only Application Insights log exporter: the former Azure Monitor log exporter is removed. `Telemetry:OtlpLogsEnabled=false` disables just OTLP logs when a collector would duplicate direct sinks. `writeToProviders` preserves Aspire logging without another console provider. The existing Bicep supplies the Application Insights connection string; Datadog secret/site configuration is an operator prerequisite, not provisioned by the current template. The standalone Aspire Dashboard is an in-memory OTLP sink on `http://localhost:18888`. The local `http-otel` launch profile sends to port 18889; no database initialization is needed for logging changes.

REDUNDANCY WARNING

Azure stdout collection may store the same logs again in Log Analytics or forward them to Datadog. Choose one collection path or disable `AppLogging:ConsoleEnabled`. Do not add another Application Insights logging provider or auto-instrumentation agent. Exceptions can legitimately exist as both span events and exception logs. Trace sampling does not sample Serilog logs. Datadog receives logs only, not APM traces; linking to APM needs compatible trace ingestion and field mapping. Startup warns about missing credentials, Azure console collection and simultaneous OTLP log export.

Two privacy switches default to off because the database is a copy of production: `Telemetry:RecordSqlText` (statement text on database spans) and `Telemetry:RecordAiContent` (prompts and completions on chat spans). Span attributes otherwise carry only counts and lengths: prompt length, parameter count, row count, truncation, source and citation counts.

These switches do not redact log messages. EF Database.Command logs (including failed-command SQL) default to a Serilog `Fatal` threshold so statement text is not copied into files or cloud logs. Other EF/circuit errors remain visible. General exception messages and explicit application properties can still contain sensitive data; do not log credentials, invoice content or prompts. No additional Serilog request-logging middleware is installed because the existing request traces already record timings.

The instrumentation calls in the AI services are `ActivitySource.StartActivity` and counter increments; when no listener is attached they return null and do nothing, which is also why the unit tests need no telemetry setup.

A typical trace for one AI query, as shown by the Aspire Dashboard:

```
Event onclick -> Radzen.Blazor.RadzenButton.OnClick      (Blazor, the user's click)
├─ ai.query.generate                                     (own span: prompt length, parameter count, model error
flag)
│   └─ chat                                              (Microsoft.Extensions.AI: model id, token usage)
└─ ai.query.execute                                     (own span: row count, truncated)
    └─ SELECT Invoices Payers                          (SqlClient: db.query.summary, no statement text)
```

The hub call that delivered the click is not the parent: Blazor dispatches the event handler after the hub method returns, so the `Event` span is a trace root of its own, and the SignalR hub-method spans are dropped by the processor.

4 Data model

The EF Core model is scaffolded from the local database and never edited by hand (the re-scaffold command is in the README). Anything that must survive a re-scaffold lives in partial classes.

`db/001_schema.sql` is the committed definition of that local database. It carries the nine tables below plus the seven more they reach through a foreign key - `AspNetUsers`, `CancellationCategories`, `CommercialManagers`, `FundingElements`, `ProgramTypes`, `SamplesPurposes` and `Vendors` - sixteen in all. The production database has around eighty; the rest play no part in this POC and are deliberately absent, which is why a scaffold run against a demo database and one against a full export produce the same model.

Tables in the model (nine of the sixteen in the schema):

TABLE	ROLE IN THE UI	SOFT DELETE	AUDIT COLUMNS
Invoices	Full CRUD, tabbed dialog	IsDeleted (bit)	yes
Bottlers	CRUD (SAP-sourced ids)	no	yes
Payers	CRUD (SAP-sourced ids)	no	yes
SalesCenters	CRUD (SAP-sourced ids)	no	yes
Programs	CRUD	IsDeleted (nullable bit)	yes
Currencies	CRUD	no	yes
Companies, Countries, BrandSegmentCategories	Dropdown sources only	no	yes

Domain hierarchy: Bottler (the "customer") owns Payers; Payer owns Sales Centers; Programs belong to a Bottler. An Invoice references all of these plus Currency, Company and Country, and may point at a previous revision of itself (`RevisedInvoiceId`). `Invoices.UserId` is a required foreign key to `AspNetUsers`, a table this app does not manage. New invoices therefore default to the submitter of a recent invoice.

Partial-class pattern. `Data/Entities/Partials/*.cs` add `[NotMapped]` members to scaffolded entities:

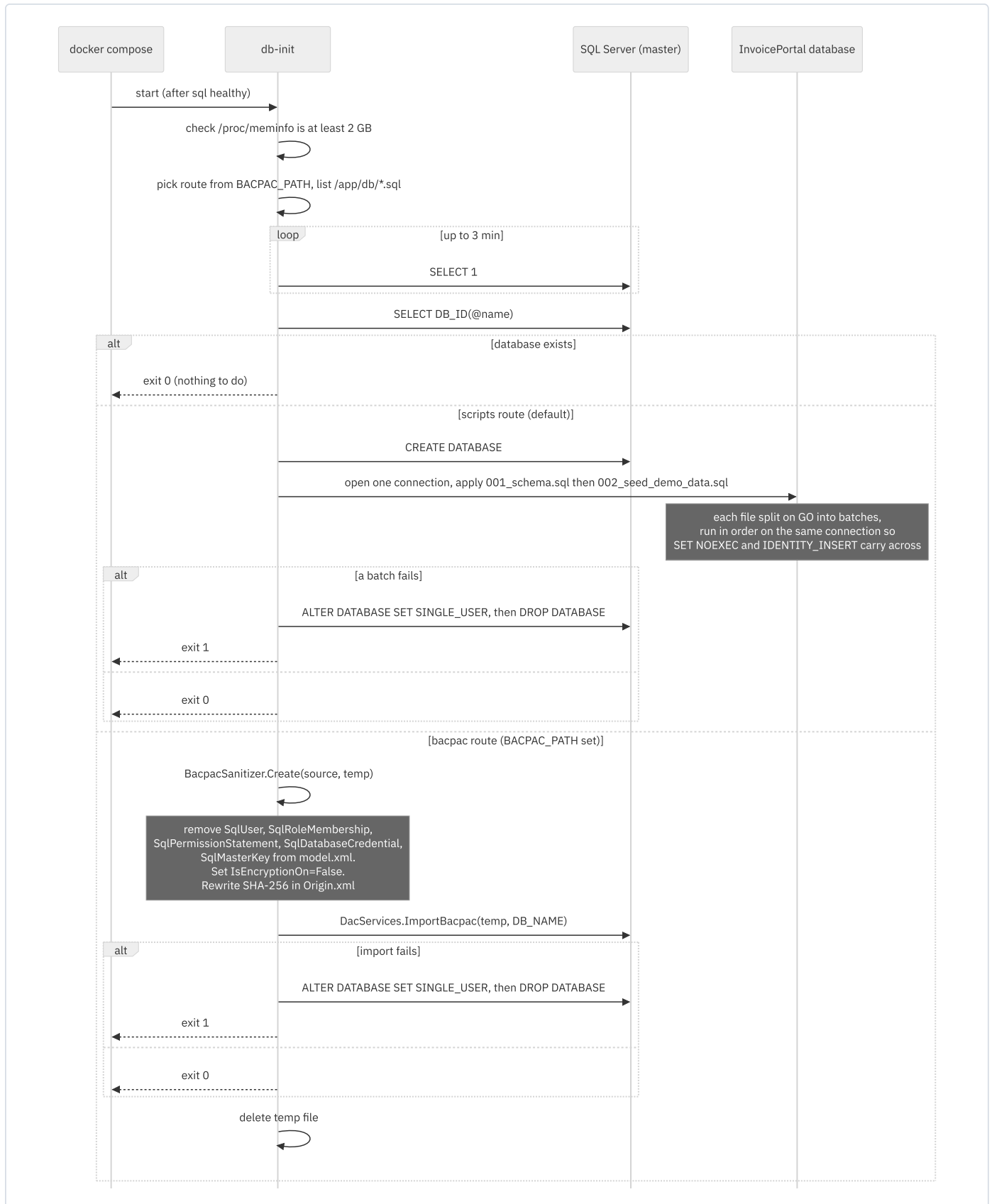
- Marker interfaces `ISoftDeletable` and `IAuditable` (in `Data/Abstractions`) so `CrudService<T>` can treat soft delete and `UpdatedAt` stamping generically.
- Typed enum views over `int` columns, for example `Invoice.Status` over `Invoice.InvoiceStatus`, using enums copied from the Invoice Portal backend (`Data/Enums`).
- `DisplayName` helpers for dialogs and confirmations.

`SoftDelete.NotDeleted<T>()` builds the filter expression against the real mapped `IsDeleted` column (handling both `bit` and nullable `bit`) so EF can translate it to SQL.

5 Data flows

5.1 Database provisioning (db-init, runs once)

`db-init` has two routes. Which one runs depends on a single environment variable: `BACPAC_PATH`. Unset (the default) means build from the committed scripts, and nothing outside the repository is needed.

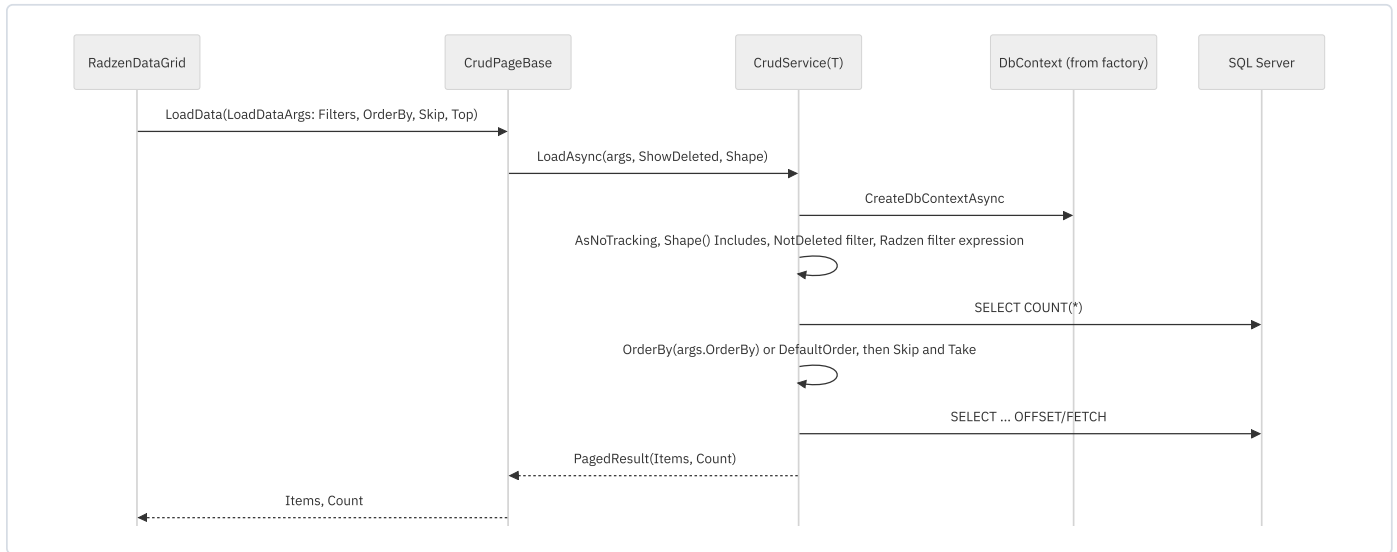


Scripts route. `SqlScriptRunner` lists `/app/db/*.sql` in ordinal filename order, splits each file on lines containing only `GO`, and executes the batches on one connection. The single connection matters: the seed script relies on session state that does

not survive a reconnect. The `db/` folder is copied into the `db-init` image at build time, so the container carries its own copy and no bind mount is involved. Building the database takes about a second.

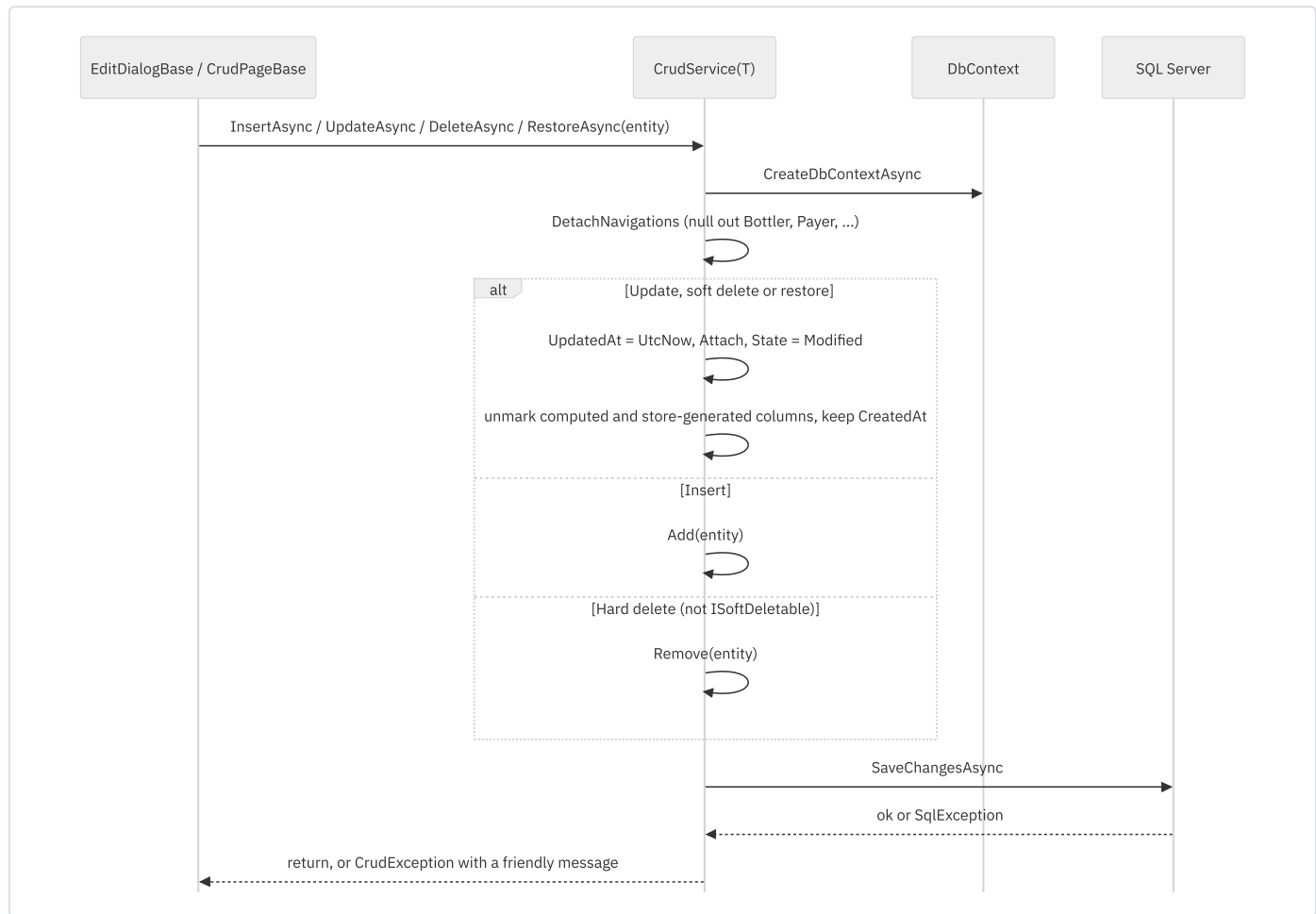
Bacpac route. The original bacpac is never modified. Azure SQL bacpacs contain Entra ID users, a master key, a scoped credential and the TDE flag, none of which an on-premises SQL Server can create, so the sanitiser strips them from `model.xml` and recomputes the checksum DacFx verifies. Import takes roughly a minute. `docker-compose.bacpac.yml` is the override that sets `BACPAC_PATH` and mounts the file.

5.2 Grid load (server-side paging, sorting, filtering)



Radzen's `LoadDataArgs` carries the grid state. Filtering uses Radzen's expression-based `QueryableExtension` (no Dynamic LINQ), so everything is translated to SQL. The Invoices page adds an `Include` for Bottler, Payer, Sales Center and Currency and an optional type filter inside `Shape`.

5.3 Create, update, delete, restore



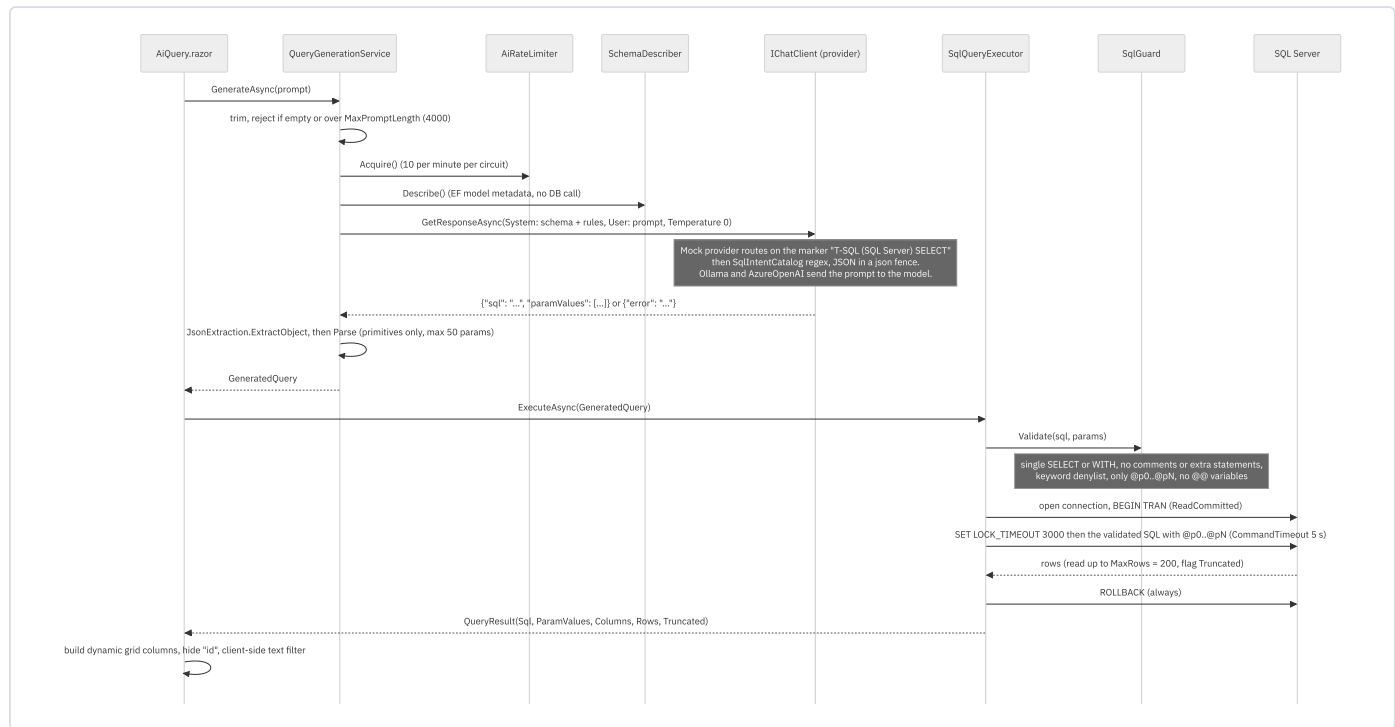
`DetachNavigations` matters because grid rows arrive with navigation objects loaded; attaching them would make EF try to track or insert the related lookup rows. SQL error numbers 2601/2627 (unique), 547 (foreign key) and 515 (not null) are translated into user-readable messages.

The edit dialog itself loads the entity by key on open, seeds defaults through `CreateNew()`, loads dropdowns through `OnModelLoadedAsync()`, and runs `ValidateAsync()` (for example, duplicate SAP id checks) before saving. Cascading dropdowns (Bottler → Payers → Sales Centers, Bottler → Programs) are reloaded from `LookupService` as the parent selection changes.

5.4 Lookup cache

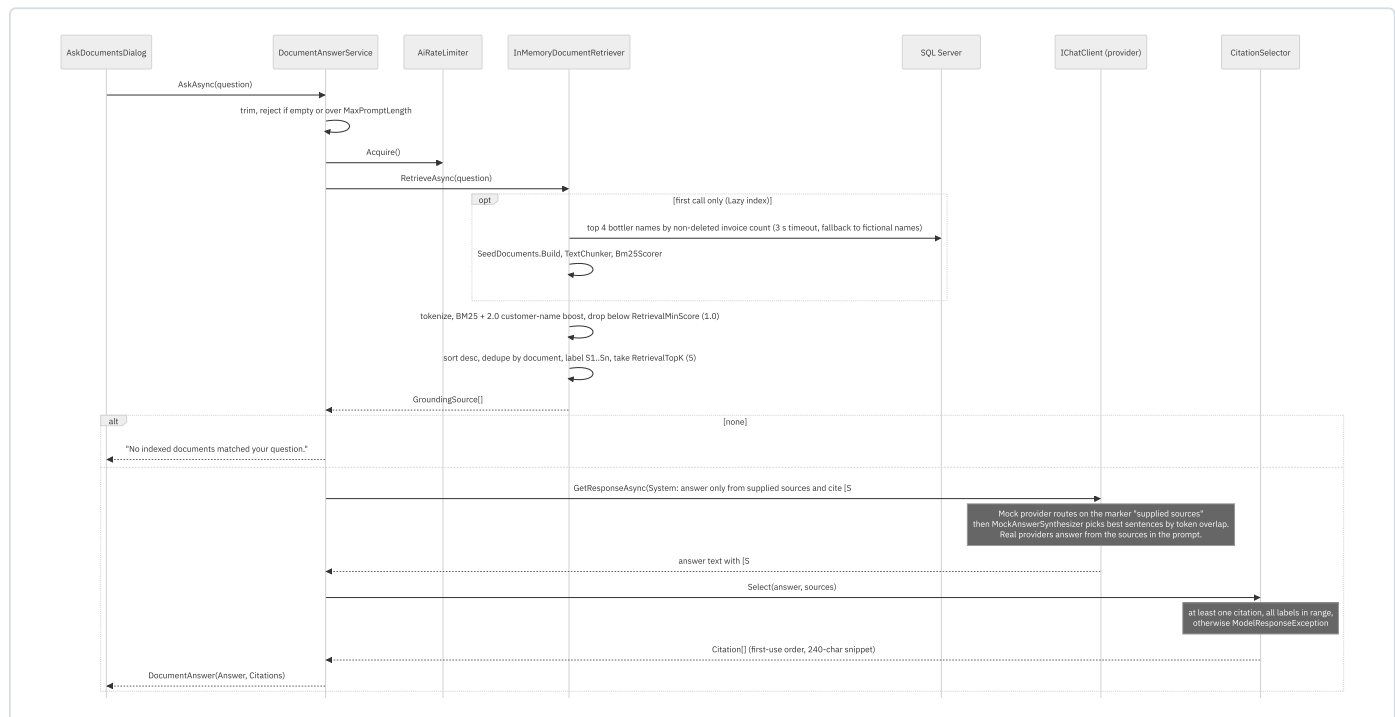
`LookupService` runs small projection queries (`Id + " - " + Name`) against the lookup tables and caches each result per circuit for 30 seconds, keyed by query name and parent id. `Invalidate()` clears the cache. It also reads the 50 most recent distinct invoice submitters to satisfy the `UserId` constraint.

5.5 AI Insights: natural language to SQL



The prompt, JSON contract parsing, guard, and execution are the same for every provider. With **Mock** the model is fake, and it deliberately wraps its JSON in a code fence so the real extraction path is exercised. With **Ollama** or **AzureOpenAI** free-form questions work, and SQL the guard rejects is shown as an error rather than executed.

5.6 Ask the documents: grounded Q&A with citations



Entry points: the header button in `MainLayout` (opens with the retriever's default question) and the per-row button on the Bottlers page (pre-fills "What do the documents say about {bottler}?").

6 Database integration points

All app-side access uses one connection string, `ConnectionStrings:InvoicePortal`, supplied as an environment variable in compose and from `appsettings.Development.json` for host-side runs. The login is `sa` with `Encrypt=True;TrustServerCertificate=True`.

COMPONENT	MECHANISM	WHAT IT DOES
<code>InvoicePortal.DbInit</code>	<code>Microsoft.Data.SqlClient</code> to <code>master</code> , then to the new database; DacFx on the bacpac route	Readiness poll, <code>DB_ID</code> existence check, then either <code>CREATE DATABASE</code> plus the <code>db/*.sql</code> batches or <code>ImportBacpac</code> ; drops the database on failure. Reads <code>SQL_SERVER</code> , <code>SQL_PORT</code> , <code>DB_NAME</code> , <code>MSSQL_SA_PASSWORD</code> , <code>SCRIPTS_PATH</code> , <code>BACPAC_PATH</code> .
<code>CrudService<T></code>	EF Core via <code>IDbContextFactory</code>	Paged list, find, exists, count, insert, update, soft/hard delete, restore. <code>AsNoTracking</code> for reads.
<code>LookupService</code>	EF Core via factory	Projection queries for dropdowns, cached 30 s per circuit.
<code>InMemoryDocumentRetriever</code>	EF Core via factory, once	Loads four bottler names to fill document placeholders.
<code>SqlQueryExecutor</code>	ADO.NET on the EF connection (<code>GetDbConnection</code>)	Runs guarded, parameterised, model-generated SELECTs inside a transaction that is always rolled back.
<code>SchemaDescriber</code>	EF model metadata only	No connection. Builds the schema block of the NL-to-SQL prompt from <code>IModel</code> .

Resilience and safety settings:

- `EnableRetryOnFailure(3)` on the EF provider for transient errors.
- Generated SQL: 5 s command timeout, `SET LOCK_TIMEOUT 3000`, 200-row cap, `ReadCommitted` transaction rolled back in `finally`. SQL Server has no engine-level read-only transaction, so protection is guard-based. For anything beyond a local POC the executor should use a separate `db_datareader` login.
- Column types returned by the executor are wrapped as `Nullable<T>` so null cells do not break Radzen's dynamic sort expressions.

7 AI service integration points

The AI slice is designed around two seams. Everything else (prompts, parsing, guards, citation validation, rate limiting, UI) is provider-agnostic.

7.1 The two seams

SEAM	INTERFACE	IMPLEMENTATIONS
Chat model	<code>Microsoft.Extensions.AI.IChatClient</code>	<code>MockChatClient</code> (in-process), OllamaSharp's <code>OllamaApiClient</code> (local model), Azure OpenAI's chat client via <code>.AsIChatClient()</code>
Document retrieval	<code>Ai/Documents.IDocumentRetriever</code>	<code>InMemoryDocumentRetriever</code> only. A real knowledge base (Azure AI Search / Foundry IQ) would be a second implementation.

Registration lives in `Ai/AIServiceCollectionExtensions.cs` and switches on `Ai:Provider`. `AiOptions` validates the value at start-up against the three registered names, so a typo fails fast rather than silently falling back to the mock.

7.2 Chat providers

PROVIDER	CLIENT AND PIPELINE	NETWORK	CREDENTIALS
Mock	<code>MockChatClient</code> → <code>UseLogging</code>	None	None
Ollama	<code>OllamaApiClient(HttpClient, model)</code> → <code>ModelTransportChatClient</code> → <code>ConfigureOptions(num_ctx)</code> → <code>UseLogging</code>	<code>Ai:Ollama:Endpoint</code> (host GPU install or the <code>ollama</code> compose service)	None
AzureOpenAI	<code>AzureOpenAIClient(endpoint, DefaultAzureCredential).GetChatClient(deployment).AsIChatClient()</code> → <code>ModelTransportChatClient</code> → <code>UseLogging</code>	<code>Ai:AzureOpenAI:Endpoint</code>	Entra ID token: managed identity in Azure, <code>az login</code> on a laptop. The resource has local keys disabled.

`ModelTransportChatClient` is a `DelegatingChatClient` that converts transport failures into `ModelResponseException` with an actionable message: endpoint unreachable, model not pulled, HTTP 401/403/404/429 from Azure OpenAI, or a timeout. The pages already catch `AIException`, so a misconfigured provider shows a readable error rather than a generic one. The Ollama pipeline also sets `num_ctx` from `Ai:Ollama:ContextLength` because Ollama's default context window is too small for the schema prompt plus a reply.

7.3 How the mock decides what to do

`MockChatClient` inspects the system prompt for a marker string:

- "T-SQL (SQL Server) SELECT" (from `QueryGenerationService.SystemPromptMarker`) → `SqlIntentCatalog`, a first-match-wins table of about a dozen regex intents that produce parameterised T-SQL in the exact JSON shape the real prompt demands. Unmatched prompts return `{"error": ...}`.
- "supplied sources" (from `DocumentAnswerService.SystemPromptMarker`) → `MockAnswerSynthesizer`, which parses the sources JSON out of the user message and stitches the most relevant sentences, citing `[S#]`.

A real model never sees these markers as anything special; it simply follows the instructions.

7.4 Prompt contracts (provider-independent)

NL-to-SQL. System prompt = schema block from `SchemaDescriber` + domain notes (soft-delete filters, hierarchy, status codes, bracketed `[From]` / `[To]` columns) + rules. Required reply: `{"sql": "<T-SQL>", "paramValues": [<primitives>]}` or `{"error": "<why>"}`. Parsed by `JsonExtraction` (tolerates fences and prose) and `QueryGenerationService.Parse` (primitives only, at most 50 parameters).

Grounded Q&A. System prompt instructs the model to answer only from the supplied sources, treat source text as data rather than instructions, and cite every statement with `[S#]`. User message = `Question: ...` followed by `Sources (JSON):` and an array of `{Label, Title, CustomerName, SourcePath, Content}`. `CitationSelector` rejects answers with no citation or an out-of-range label.

7.5 Configuration

Bound from the `Ai` section (`appsettings.json`), overridable with `Ai_*` environment variables.

KEY	DEFAULT	EFFECT
<code>Enabled</code>	<code>true</code>	Hides the AI Insights nav item, the header button and the per-row button; <code>/ai/query</code> shows a warning.
<code>DocumentChatEnabled</code>	<code>true</code>	Gates the "Ask the documents" dialog only.
<code>Provider</code>	<code>Mock</code>	<code>Mock</code> , <code>Ollama</code> or <code>AzureOpenAI</code> . Anything else fails validation at start-up.
<code>Ollama:Endpoint</code>	<code>http://localhost:11434</code>	Ollama base URL. Compose overrides it to <code>http://host.docker.internal:11434</code> .
<code>Ollama:Model</code>	<code>qwen2.5-coder:7b</code>	Model tag. Must already be pulled.
<code>Ollama:TimeoutSeconds</code>	<code>120</code>	Per-call HTTP timeout. CPU inference can take tens of seconds.
<code>Ollama:ContextLength</code>	<code>8192</code>	Sent as <code>num_ctx</code> . The schema prompt needs more than Ollama's default.
<code>AzureOpenAI:Endpoint</code>	<code>empty</code>	Required when the provider is <code>AzureOpenAI</code> . Set by the Bicep template in Azure.
<code>AzureOpenAI:Deployment</code>	<code>gpt-4.1-mini</code>	Deployment name on the resource.
<code>MaxPromptLength</code>	<code>4000</code>	Prompt / question length cap.
<code>RateLimitPerMinute</code>	<code>10</code>	Fixed window per circuit, shared by both AI features.
<code>MaxRows</code>	<code>200</code>	Row cap on generated SQL results.
<code>CommandTimeoutSeconds</code>	<code>5</code>	Command timeout for generated SQL.
<code>RetrievalTopK</code>	<code>5</code>	Maximum grounding sources after dedupe.
<code>RetrievalMinScore</code>	<code>1.0</code>	BM25 threshold, analogue of a reranker threshold.

7.6 Adding another provider

1. Add the provider's `IChatClient` package.
2. Add a `case` in `AddInvoicePortalAi` that calls `AddChatClient(...)` , wraps it in `ModelTransportChatClient` so transport errors stay readable, and keeps `.UseLogging()` .
3. Add the name to the `Provider` regular expression in `AiOptions` and to `AiOptionsTests` .

4. Optionally register a real `IDocumentRetriever` that calls a knowledge base and maps hits to `GroundingSource` (deduplicated, labelled `S1..Sn`).

Nothing in the pages, services, guards or the rest of the tests needs to change. Any non-mock provider sends the schema (NL-to-SQL) and document content (Q&A) to the model endpoint, so review where that endpoint is before enabling it outside the local machine.

8 Configuration and environments

SETTING	COMPOSE (APP)	HOST-SIDE DEVELOPMENT
ASPNETCORE_ENVIRONMENT	Production	Development (launch profile http)
ConnectionStrings__InvoicePortal	env var, Server=sql,1433	appsettings.Development.json , localhost,1433
HTTP port	8080	5098 (launch profile http ; .vscode/launch.json for F5)
Ai__Enabled , Ai__DocumentChatEnabled	env vars in docker-compose.yml	appsettings.json defaults
Ai__Provider , Ai__Ollama__Endpoint , Ai__Ollama__Model	from .env (AI_PROVIDER , AI_OLLAMA_ENDPOINT , AI_OLLAMA_MODEL) with Mock / host Ollama defaults	appsettings.json defaults, or Ai__Provider set in the shell
SQL SA password, DB name	.env (MSSQL_SA_PASSWORD , DB_NAME), read by compose	same password hard-coded in appsettings.Development.json
SCRIPTS_PATH , BACPAC_PATH (db-init)	SCRIPTS_PATH=/app/db in docker-compose.yml ; BACPAC_PATH unset unless docker-compose.bacpac.yml is layered on	defaults to /app/db ; set BACPAC_PATH to import a bacpac instead
OTEL_EXPORTER_OTLP_ENDPOINT , Telemetry__RecordSqlText	from .env (OTEL_EXPORTER_OTLP_ENDPOINT , TELEMETRY_RECORD_SQL_TEXT), empty / false by default	Telemetry section in appsettings.json , or OTEL_EXPORTER_OTLP_ENDPOINT set in the shell (http://localhost:18889 for the dashboard)
APPLICATIONINSIGHTS_CONNECTION_STRING	not set	not set; set by the Bicep template in Azure
AppLogging__RunningInAzure	unset, auto-detects as local despite Production	unset, auto-detects as local; override for other Azure hosts
AppLogging__FilePath	/app/logs/invoiceportal-.json by default, mount a writable volume for persistence	logs/invoiceportal-.json under content root
AppLogging__FileSizeLimitBytes , AppLogging__RetainedFileCountLimit	10485760 bytes, 14 files, daily and size rolling	same defaults; not necessarily 14 days
DD_API_KEY , DD_SITE	ignored for local logging	used only in Azure; site defaults to datadoghq.com , key must be a secret reference
Telemetry__OtlpLogsEnabled	true when OTLP configured	true, keeps Aspire structured logs
AppLogging__RequestLoggingEnabled	true: one log record per HTTP request	same default

Compose reads `.env` , which is git-ignored. `.env.example` is tracked and is the template a fresh clone copies to `.env` ; without it compose substitutes a blank SA password and `sql` never starts. The SA password there is a throwaway local credential and deliberately matches the one in `appsettings.Development.json` , so host-side runs reach the same container.

Azure detection checks `CONTAINER_APP_NAME` , `WEBSITE_INSTANCE_ID` , or `WEBSITE_SITE_NAME` ; neither `Production` nor cloud credentials imply Azure. For Azure VM/AKS hosting use the explicit override.

`AppLogging__ApplicationInsightsEnabled` and `AppLogging__DatadogEnabled` can disable cloud sinks separately. `AppLogging__DatadogApiKey` and `AppLogging__DatadogSite` override the standard Datadog keys. Datadog's supported sites and secure configuration steps are in the README logging section. Use secret/Key Vault references, never source-controlled keys. Missing credentials disable just that sink with a startup warning. Optional `DD_SERVICE` , `DD_ENV` , `DD_VERSION`

override Datadog tags. Configure levels under `Serilog:MinimumLevel`, not the former `Logging:LogLevel`. `Serilog:WriteTo` / `Serilog:AuditTo` are rejected at startup so configuration cannot bypass the environment-based sink selection.

Local log folders are git/Docker/publish-excluded. Custom paths need equivalent exclusions and permissions. Datadog is batched with a 10,000-event queue; remote delivery is best-effort, not a durable audit trail. Sink shutdown flushes normally, but force-kills, network failures, queue limits, or container replacement can lose logs. Local files roll after the threshold is reached, so a single oversized event can exceed it.

Azure hosting. `azure.yaml` and `infra/*.bicep` describe an Azure Developer CLI deployment to Container Apps (one replica, sticky sessions and WebSockets for the Blazor circuit). The template provisions a container registry, a user-assigned managed identity, a storage account for the Data Protection key ring, Log Analytics and Application Insights, and optionally an Azure OpenAI resource with a `gpt-4.1-mini` deployment. It does not create or restore a database: the container is pointed at the existing Azure SQL database, and the managed identity is granted access with `infra/sql/create-app-user.sql`. The template sets `DataProtection__BlobUri`, `Ai__Provider=AzureOpenAI` and `Ai__AzureOpenAI__Endpoint` on the container, and can enable Entra ID sign-in (Easy Auth) when `ENTRA_CLIENT_ID` is supplied. `.github/workflows/azure-dev.yml` runs `azd provision` and `azd deploy` on pushes to `main` using OIDC; `ci.yml` builds, tests and validates Bicep on pull requests. Step-by-step instructions are in `docs/azure-deploy.md`.

ACTION ITEM

The `.env.Development` file in the repo root belongs to a different application (the Invoice Portal API). No code in this solution reads it. It contains credentials for external Azure resources and should be removed from the working tree or rotated. It is covered by `.gitignore` but still present on disk.

`RequiresAspNetWebAssets=true` in the Admin project forces the SDK to restore the framework's Blazor web assets during the Docker build, which restores from the `.csproj` alone. Without it the published container returns 404 for `blazor.web.js` and no circuit starts.

9 Security and safety posture

- **Network:** with the default `Mock` provider there are no outbound calls. `Ollama` talks to a local endpoint (host or compose sidecar). `AzureOpenAI` sends prompts, which include the schema and document content, to the configured Azure OpenAI endpoint over HTTPS with an Entra ID token. The local listeners are the app on 8080, SQL Server on 1433 and, with the profile, Ollama on 11434.
- **Identity:** locally there is no authentication or authorization, and the app uses the `sa` login. In Azure, Easy Auth with Entra ID is optional and the app reaches SQL, Blob Storage, the registry and Azure OpenAI through a managed identity with no stored keys. `/healthz` is unauthenticated in both cases.
- **Generated SQL:** defence in depth through `SqlGuard` (single SELECT, denylist, parameter naming, no comments or system variables), a short command and lock timeout, a row cap and an unconditional rollback. The guard is a regex allow/deny list, not a parser, so a dedicated read-only login is the recommended next layer.
- **Prompt injection:** the Q&A system prompt tells the model to treat source text as data. Citation validation ensures the answer can only reference supplied sources. With a real model this remains a mitigation, not a guarantee.
- **Data:** the repository is public and everything committed under `db/` is synthetic - invented for the demo, not sampled or anonymised from any real system, with `example.com` addresses throughout. A bacpac, on the optional route, is a copy of production data including user emails; it lives only in the Docker volume and the bind-mounted file, both excluded from git by the `*.bacpac` rule.
- **Telemetry/logging:** local defaults write console and rolling files but do not export remotely. OTLP, Application Insights and Azure Datadog require configured destinations/credentials. SQL/prompt content is excluded from spans unless the telemetry privacy switches are enabled. Log messages and exceptions are not automatically redacted; EF SQL-command logs are separately suppressed by default. Restrict file access and cloud retention, avoid sensitive properties, and account for third-party Datadog egress.

10 Testing

InvoicePortal.Admin.Tests (xUnit) covers AI, telemetry and logging without a database:

TEST CLASS	COVERS
SqlGuardTests	Accepts single parameterised SELECT/WITH; rejects DML, DDL, comments, multiple statements, bad parameter names, @@ variables.
QueryGenerationServiceTests	JSON contract parsing, fenced JSON, error objects, parameter limits, length and rate-limit rejections (via StubChatClient).
MockChatClientContractTests	The mock honours the same output contract a real model is instructed to follow.
InMemoryDocumentRetrieverTests	Ranking, customer boost, threshold, dedupe, labelling, TopK, fallback names.
CitationSelectorTests	Cited-only selection, ordering, rejection of missing or unknown labels.
AiOptionsTests	Only Mock , Ollama and AzureOpenAI pass validation; Ollama defaults point at a local coder model with a large enough context.
TelemetryOptionsTests	Defaults export nothing and record no SQL text or prompt content; OTLP protocol and sampling ratio validation; health probe and static assets are excluded from request tracing.
AppLoggingTests	Azure detection/override, credential-gated sink selection, trusted Datadog sites, startup warnings, Activity-to-Application Insights correlation, real local file rolling/retention, and once-only OTLP log forwarding with scopes and trace ID. No cloud calls.

Run with:

```
dotnet test InvoicePortal.slnx
```

CRUD pages and services are exercised manually against the container. Logging tests construct an in-process host with temporary files and an in-memory OpenTelemetry exporter; cloud delivery requires configured credentials and an environment smoke test, and is not claimed by offline tests.

11 Known limitations and extension points

- **Scope:** nine tables in the UI, sixteen in the committed schema. The production database has around eighty; attachments, payments, statements, workflow history and the rest are absent, and the tables that remain (users, cancellation categories, samples purposes) are visible only as foreign-key ids.
- **Demo data:** `db/002_seed_demo_data.sql` is a generator, not a capture. It produces plausible shapes and volumes, but the distributions are arithmetic rather than observed, so it will not reproduce a performance characteristic or a data-quality problem seen in production. Use a real bacpac for that.
- **Invoices.UserId** : satisfied by borrowing a recent submitter because `AspNetUsers` is out of scope.
- **Mock model:** understands only the regex intents in `SqlIntentCatalog` ; everything else returns the fallback error. Real providers remove that limit but add latency (seconds on a GPU, tens of seconds on CPU for Ollama) and, for Azure OpenAI, cost and egress.
- **Document index:** always the six seeded documents, whichever chat provider is active. A real knowledge base needs a second `IDocumentRetriever` .
- **Lookup cache:** per circuit, time-based, not invalidated by edits from other sessions.
- **Single replica:** Blazor Server circuits are in-process, so the Azure template runs one replica with sticky sessions. Scaling out would need a SignalR backplane.
- **Extension seams:** `IChatClient` (a new `case` in the provider switch), `IDocumentRetriever` , `OnModelCreatingPartial` in the context partial, and the `CrudPageBase` / `EditDialogBase` pair for adding another table (one grid page, one dialog, one partial class).